

WEEK 2: Write a Java program to create a class called "Book" with attributes for title, author, and ISBN, and methods to add and remove books from a collection.

```
import java.util.ArrayList;

class Book {

    private String title;
    private String author;
    private String ISBN;

    public Book(String title, String author, String ISBN) {
        this.title = title;
        this.author = author;
        this.ISBN = ISBN;
    }

    public String getTitle() {
        return title;
    }

    public String getAuthor() {
        return author;
    }

    public String getISBN() {
        return ISBN;
    }

    public void displayBook() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("ISBN: " + ISBN);
    }
}

public class BookCollection {

    private ArrayList<Book> books = new ArrayList<>();

    public void addBook(Book book) {
        books.add(book);
        System.out.println("Book added successfully.");
    }
}
```

```
public void removeBook(String ISBN) {
    for (Book book : books) {
        if (book.getISBN().equals(ISBN)) {
            books.remove(book);
            System.out.println("Book removed successfully.");
            return;
        }
    }

    System.out.println("Book not found.");
}
```

```
public void displayBooks() {
    if (books.isEmpty()) {
        System.out.println("No books in the collection.");
        return;
    }

    for (Book book : books) {
        book.displayBook();
        System.out.println("-----");
    }
}
```

```
public static void main(String[] args) {
    BookCollection collection = new BookCollection();

    Book book1 = new Book(
        "Java Programming",
        "John Smith",
        "123456789"
    );

    Book book2 = new Book(
        "Object-Oriented Programming",
        "Jane Doe",
        "987654321"
    );

    collection.addBook(book1);
    collection.addBook(book2);

    System.out.println("\nBooks in Collection:");
    collection.displayBooks();
}
```

```
collection.removeBook("123456789");
```

```
System.out.println("\nAfter Removing a Book:");  
collection.displayBooks();
```

```
}
```

```
}
```